

① Tautology

A compound proposition which is always True for all possible combinations of truth values of its components is called a Tautology

② Contradiction

A compound proposition which is always False for all possible combinations of truth values of its components is called a Contradiction (absurdity)

③ Contingency

A compound proposition which is neither a Tautology nor a Contradiction is called a Contingency

④ Logical equivalence

Two propositions A and B are said

to be logically equivalent if they have the same truth values for all possible combinations of the truth values of their corresponding components and is denoted by $A \equiv B$ or $A \Leftrightarrow B$.

Logically equivalent propositions are treated as identical proposition.

⑤ Logical implication

When a hypothetical statement $p \rightarrow q$ is such q is True whenever p is True, then p logically implies q is denoted by $p \Rightarrow q$.

[$p \Rightarrow q$ means $p \rightarrow q$ is a Tautology]

Other Connectives

1. Exclusive disjunction

$P \underline{\vee} Q$ [either P or Q but not both]

P	Q	$P \underline{\vee} Q$
T	T	F
T	F	T
F	T	T
F	F	F

2. NAND ["Not P and Q "]

It is a combination of Not & and. It is denoted by symbol $\uparrow$.

$$\text{Also } P \uparrow Q = \neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$$

P	Q	$P \uparrow Q$
T	T	F
T	F	T
F	T	T
F	F	T

3. NOR ["Not P or Q "]

NOR is a combination of Not and Or. It is denoted by the symbol $\downarrow$.

ii $p \downarrow q = \neg(p \vee q) (\Leftrightarrow) \neg p \wedge \neg q$

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

The Laws of Logic

For any primitive statements p, q, r , any tautology T_0 and any Contradiction F_0

1. $\neg \neg p \Leftrightarrow p$ Law of double negation
2. $\neg (p \vee q) \Leftrightarrow \neg p \wedge \neg q$
 $\neg (p \wedge q) \Leftrightarrow \neg p \vee \neg q$ De Morgan's laws
3. $p \vee q \Leftrightarrow q \vee p$
 $p \wedge q \Leftrightarrow q \wedge p$ Commutative laws
4. $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$
 $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$ Associative laws
5. $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
 $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$ Distributive laws
6. $p \vee p \Leftrightarrow p$
 $p \wedge p \Leftrightarrow p$ Idempotent laws
7. $p \vee F_0 \Leftrightarrow p$
 $p \wedge T_0 \Leftrightarrow p$ Identity laws

$$8. \quad p \vee \neg p \Leftrightarrow T_0$$

Inverse laws

$$p \wedge \neg p \Leftrightarrow F_0$$

$$9. \quad p \vee T_0 \Leftrightarrow T_0$$

Domination

$$p \wedge F_0 \Leftrightarrow F_0$$

laws

$$10. \quad p \vee (p \wedge q) \Leftrightarrow p$$

Absorption

$$p \wedge (p \vee q) \Leftrightarrow p$$

laws